

VIPUL RATHORE

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RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Cross-Lingual Transfer, Information Extraction, Low-Resource Languages, Fairness in ML models.

EDUCATION

PhD (Computer Science & Engineering) *July 2019 - Present*

Title Integrating Large Language Models with task-specialized models for cross-lingual NLP in low-resource languages

Advisor [Prof. Mausam](#), [Prof. Parag Singla](#)

College Indian Institute of Technology Delhi

CGPA 8.73/10

MTech. (Computational and Data Sciences) *August 2017 - June 2019*

Title Procedural Information Extraction from Scientific Documents

Advisor [Partha Pratim Talukdar](#)

College Indian Institute of Science Bangalore

CGPA 8.5/10

BTech. (Electrical Engineering) *July 2012 - May 2016*

College Indian Institute of Technology Bombay

CGPA 7.8/10

PUBLICATIONS

- Vipul Rathore, Malik Hammad Faisal, Parag Singla, Mausam. “Combining Distantly Supervised Models with In Context Learning for Monolingual and Cross-Lingual Relation Extraction”. (*Under Submission*)
- Vipul Rathore, Aniruddha Deb, Ankish Chandresh, Parag Singla, Mausam. “SSP: Self-Supervised Prompting for Cross-Lingual Transfer to Low-Resource Languages using Large Language Models”. EMNLP 2024 (Findings). Miami, USA. November 2024.
- Vipul Rathore, Rajdeep Dhingra, Parag Singla, Mausam. “ZGUL: Zero-shot Generalization to Unseen Languages using Multi-source Ensembling of Language Adapters”. EMNLP 2023. Singapore. December 2023.
- Anand Brahmabhatt, Vipul Rathore, Mausam, Parag Singla. “Towards Fair and Calibrated Models”. CoRR Technical Report, 2023.
- Vipul Rathore, Kartikeya Badola, Parag Singla, Mausam. “PARE: A Simple and Strong Baseline for Monolingual and Multilingual Distantly Supervised Relation Extraction”. Annual Conference of the Association for Computational Linguistics (ACL). Dublin, Ireland. May 2022.
- Keshav Kolluru, Samarth Aggarwal, Vipul Rathore, Mausam, Soumen Chakrabarti. “IMOJIE: Iterative Memory-Based Joint Open Information Extraction”. Annual Conference of the Association for Computational Linguistics (ACL). Online. July 2020.

RESEARCH EXPERIENCE

Doctoral Researcher, IIT Delhi (2019 – Present)

- Developed HYDRA, a hybrid LLM-based distantly supervised system achieving state-of-the-art performance in English and cross-lingual relation extraction. (Under Submission)
- Created SSP, a self-supervised prompting framework leveraging LLMs for cross-lingual transfer (EMNLP 2024 Findings).
- Designed ZGUL, a multi-source language adapter ensemble achieving zero-shot generalization across diverse language families for NER & POS tasks (EMNLP 2023).
- Designed PARE, a strong baseline for Distantly Supervised Relation Extraction (ACL 2022).

Masters Research Fellow, Ericsson Research Bangalore (2018-19)

- Automated troubleshooting over Ericsson documentation using GCNs and SRL-based graphs for procedural state tracking.

TEACHING EXPERIENCE

Teaching Assistant, IIT Delhi (2019 – Present)

- Natural Language Processing: Assignment and exam design, course materials, lectures.
- Artificial Intelligence: Designed board-game assignment, ran multi-stage student tournaments.
- Data Structures & Algorithms: Developed Java interfaces, drivers, and automated testcases.

AWARDS & HONORS

- *Great Reviewer Recognition*, ACL Rolling Review (ARR), Oct 2023
- *Prime Minister's Research Fellowship* (PMRF), (2020-24)
- *Best Teaching Assistant Award*, IIT Delhi (2021)
- *All-India Rank 567*, IIT-JEE 2012 (Top 0.1)
- *KVPY Fellowship* (Young Scientific Talent Program), 2010–11
- *Ericsson Research Fellowship*, IISc Bangalore (2018–19)

SERVICE & OUTREACH

- Reviewer: JAIR (2021), ARR (2021–2023), EACL 2024, NAACL 2024, ACL 2024.
- Presented at Amazon Research Days 2024 (Bangalore).
- Organizer, IISc AI Day (2019) & IIT Bombay TechFest (2012–13).

SKILLS

- **Technical:** Python, PyTorch, Bash, C++, PEFT (LoRA/Adapters), Prompt Engineering.
- **Conceptual:** NLP, ML, LLMs, Cross-lingual Transfer, Linear Algebra, Probability, Optimization.

GRADUATE COURSEWORK

Advanced Data Structures (Grade: 10/10), Information Retrieval (Grade: 10/10), Natural Language Processing, Machine Learning, Practical Data Science, Numerical Linear Algebra, Numerical Methods (Grade: 10/10), Numerical Optimization.